

Building a Production LLM Pipeline for Ad Content Review at Scale

Lessons from Argus

Eagle Chen Team Lead, Data Science · Content Quality @ Taboola



ABOUT THE SPEAKER

Eagle Chen

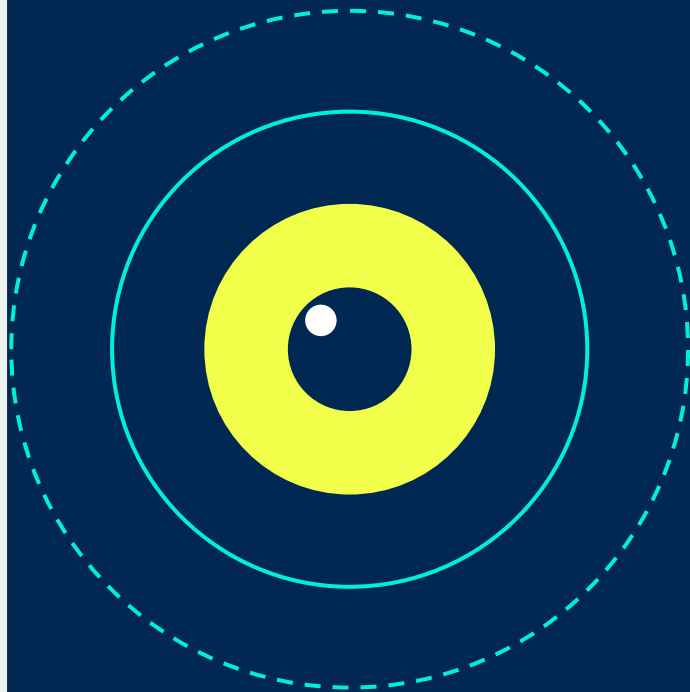
Team Lead, Data Science · Content Quality @ Taboola

EXPERIENCE

9+ years in **software engineering & AI/ML**
earlier — image processing algorithm



"ARGUS"



*the hundred-eyed
guardian of myth*

WHERE I WORK

THE GROWTH ENGINE
FOR THE OPEN WEB

**AI-driven
revenue, engagement
& audience growth**

11K+

global publishers

18K+

global advertisers

600M+

daily active users

2K+

employees (~500 R&D)

35

global offices

\$130M+

annual R&D

Performance AI

Different formats and placements

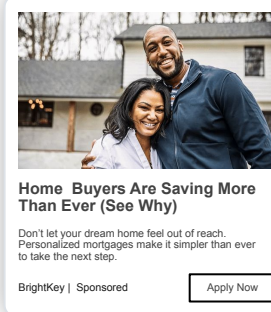
Display



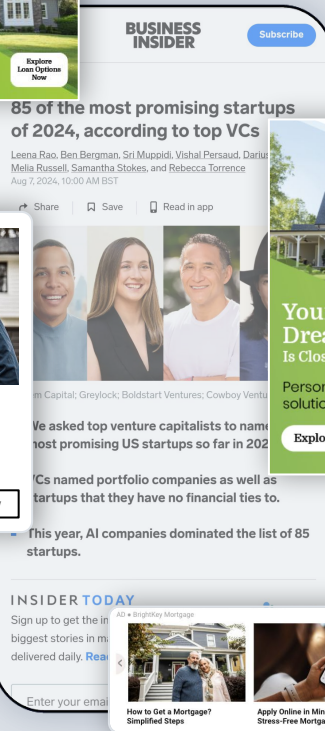
Vertical



Native Motion *



*Built with GenAI AdMaker



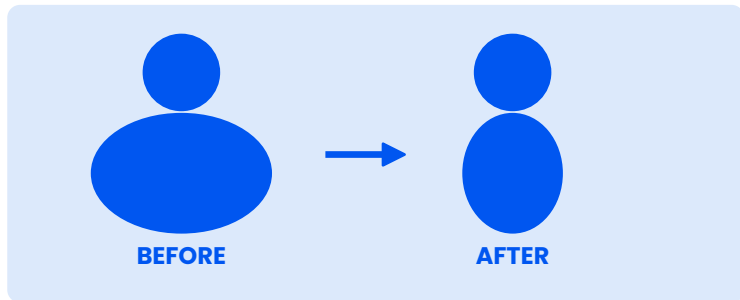
Carousel

We are the bridge — so the content has to be right



So we review every ad — before it ever serves.

Would you approve this ad?



Doctors Are Stunned: FDA-Approved Trick Melts Belly Fat Overnight

Thousands already swear by this 10-second evening ritual. Try it before it's banned.

Sponsored · health-news-daily.example

✓ APPROVE

✗ REJECT



Is the claim verifiable?



Whose image is that?



Where does the link go?

What we'll cover

01

The challenge — and how we beat it

why ad review is hard, and the bet we made

02

The architecture

how Argus is built — a harness around the model

03

Impact & Lessons

what it delivered, and what we learned

01

Two Problems, One opportunity

why human review doesn't scale — and traditional ML didn't either



1

CONSISTENCY

reviewers disagree — with each other, and with themselves

2

TIME

advertisers wait 1+ days for a decision before going live

3

SCALABILITY

unique knowledge per market × language — retrain on every policy update

We tried the classic playbook. It buckled.



A multimodal task with rich context



Labels you can't trust



Policies that never sit still

the model zoo

Safety v1

IAB cls

Ad-type

Thumb cls

Rules #1–40

Geo rules

Lang det

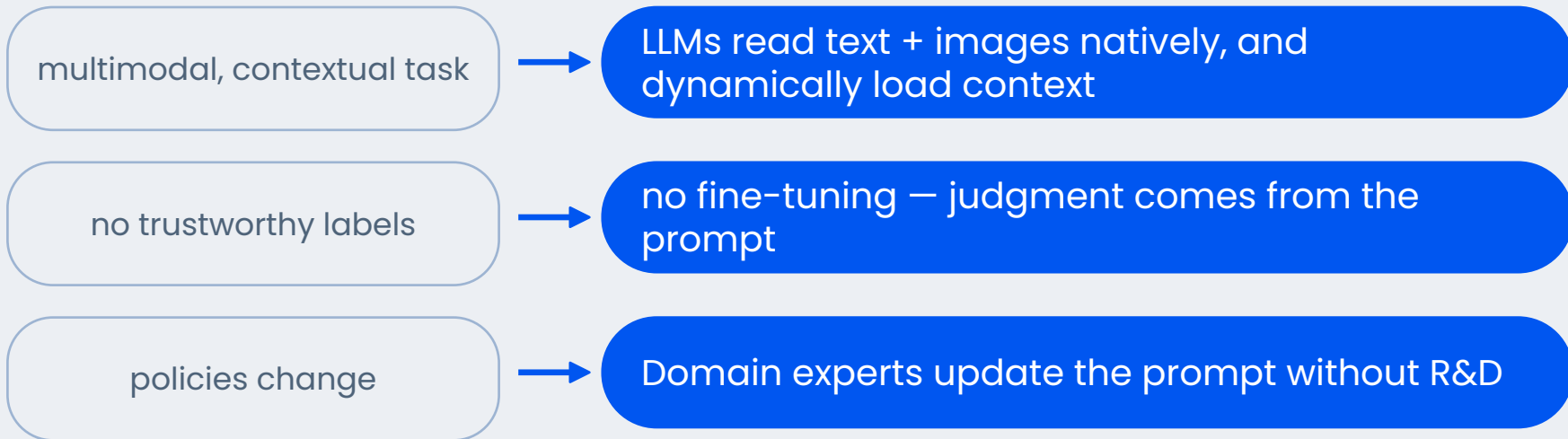
Dup match

...

maintained one by one — forever

Low AI-review coverage

From traditional ML to LLM



Policy = an English prompt

HARNESS ENGINEERING

Model is an engine; Harness is the car.



control flow

context

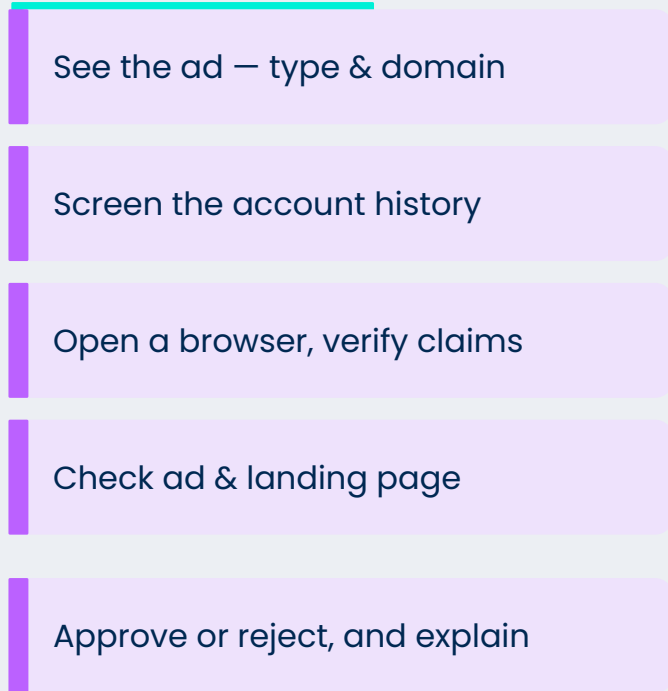
tools

memory

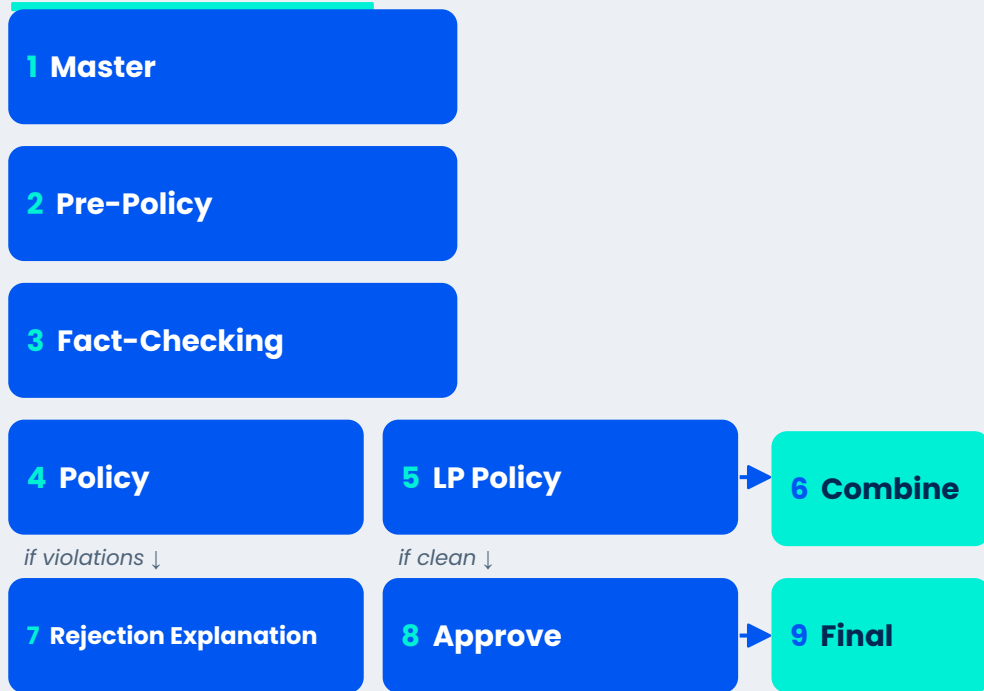
CI/CD

Nine stages — copied from the human workflow

HOW A HUMAN REVIEWS



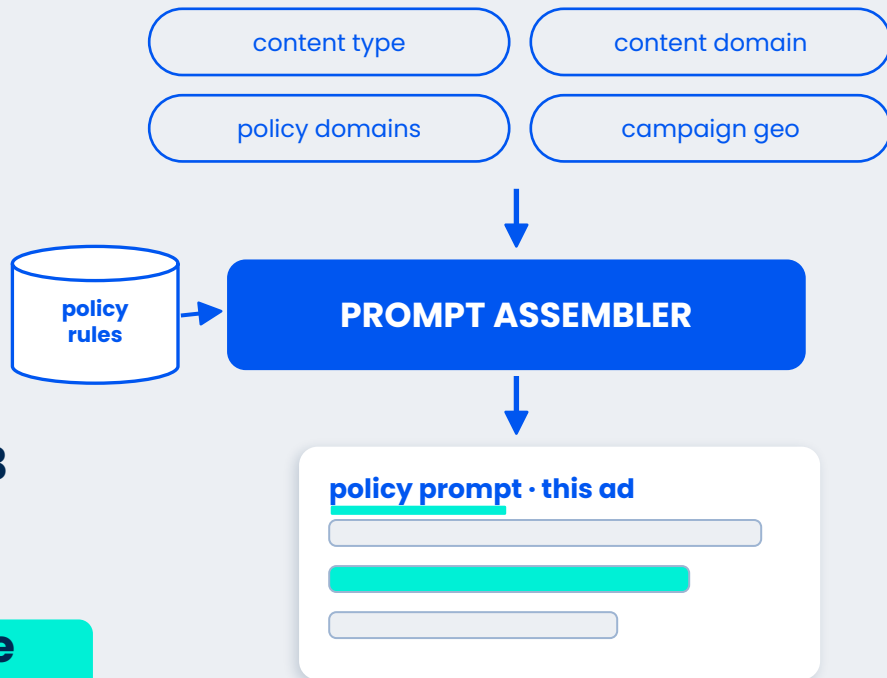
THE ARGUS PIPELINE



each stage: auditable · testable · model-swappable

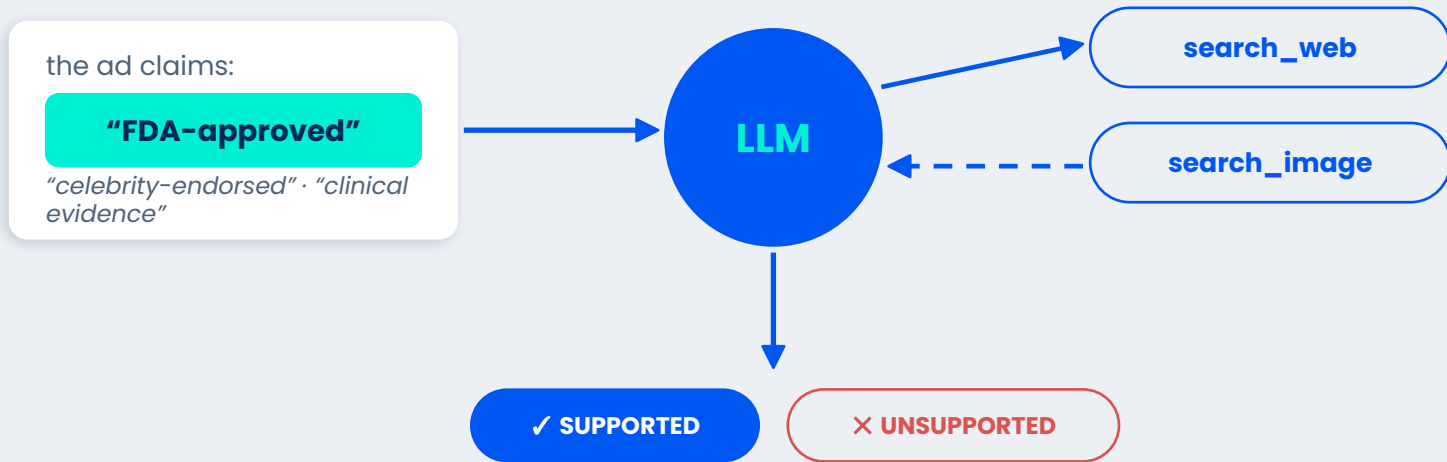
Prompts are assembled, not written

- 1 **Master's labels pick the rules**
- 2 **More context input, e.g. target market**
- 3 **Rules live in a versioned DB**



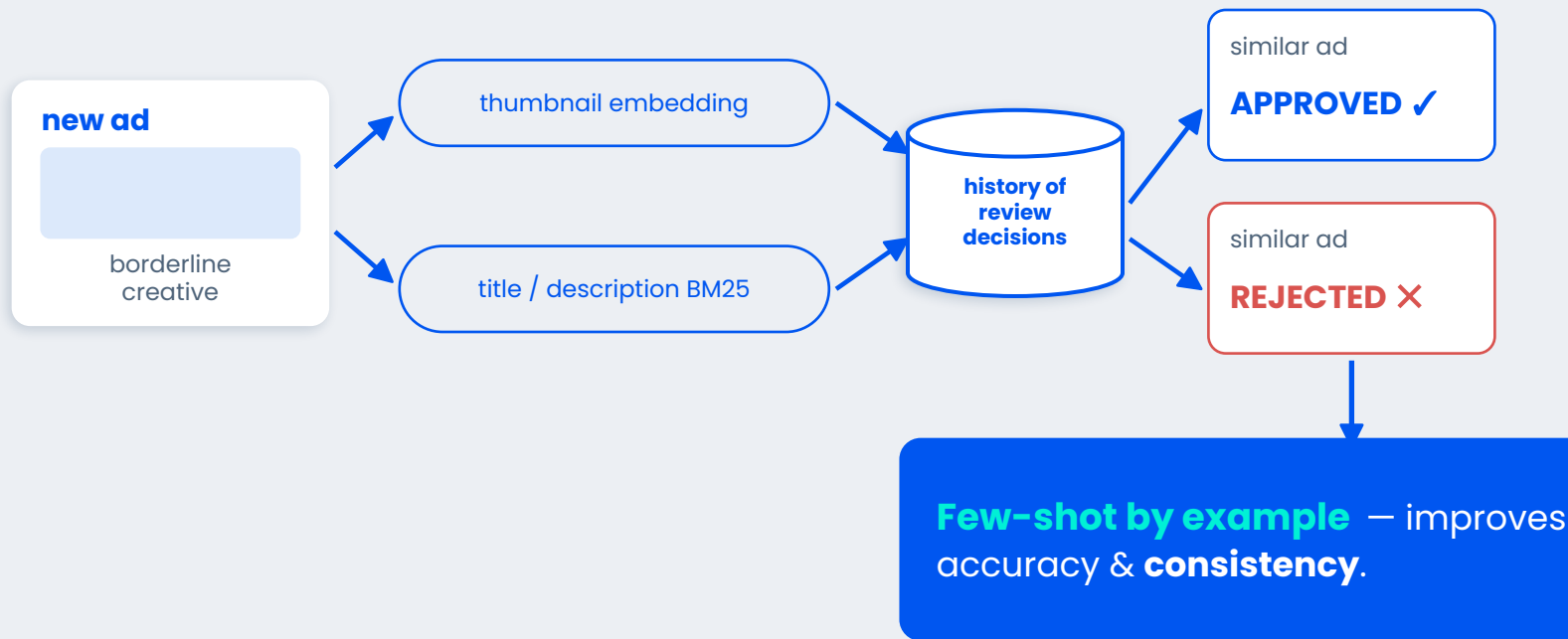
Fewer tokens — and more accurate

Agentic — exactly where judgment needs evidence



Why our own tools? They keep Argus model-agnostic — swap models freely.

In-Context Learning: institutional memory, injected at inference time



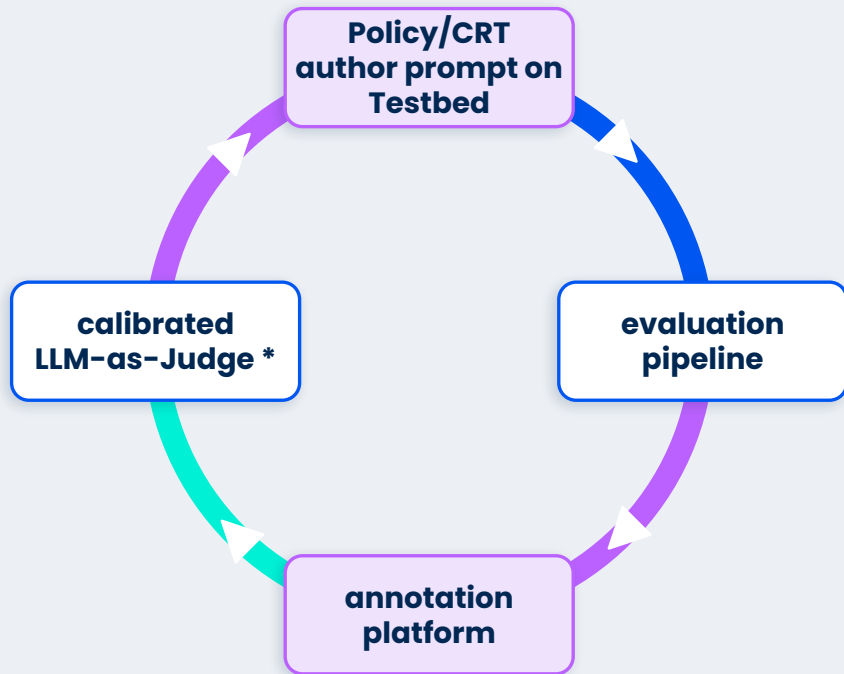
cf. Class-RAG: Real-Time Content Moderation with Retrieval-Augmented Generation — Meta, 2024 · arXiv:2410.14881

Prompts are code — CI/CD: versioned · tested · gated



Release Gate: Quality / Cost / Latency

The full CI/CD cycle: domain experts drive the iteration

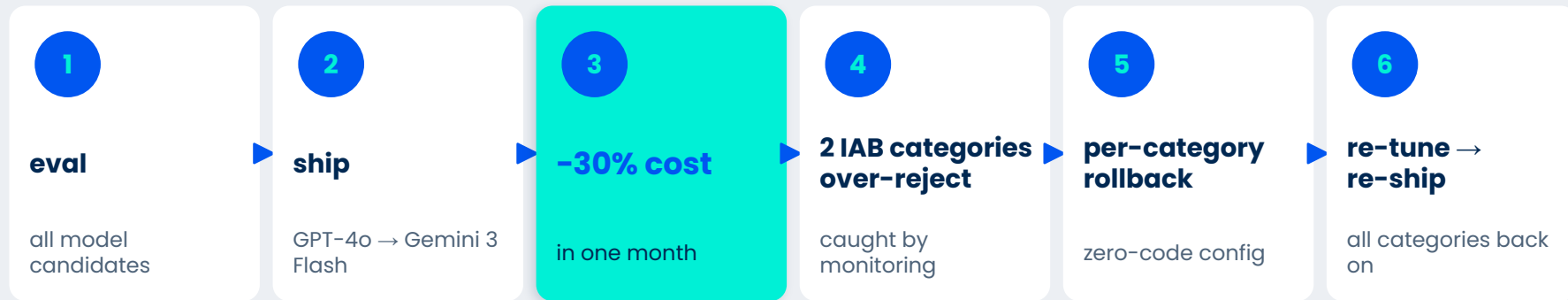


v131 policy-stage prompt
v78 policy rule set

**R&D freed from the model zoo —
domain experts now iterate at
business speed.**

** judge calibrated against human labels — bias-corrected, with confidence intervals (Lee et al, 2025)*

We swapped the engine of a moving car



Like a tire change in an F1 race

03 · BENEFITS REALIZED

12 hr → minutes

review turnaround — Argus v.s. human reviews

All English traffic

every English ad is eligible to review by Argus



75% less workload

for human reviewers

+13 pp

review consistency vs. human reviewers

Bonus: Actionable rejection comment · advertisers act on the feedback — and that shows up as a lift vs. manual review.

~2×

vs manual

more rejections resubmitted

1.5×

vs manual

more eventually re-approved

1

Encode the human workflow

reliability, auditability, and trust comes free

2

Let domain experts drive the harness

reviewers become AI mentors, or freed to serve customers

3

The CI/CD layer.

the key to make R&D and domain experts collaborate smoothly

Thank you

Q&A → find me in the hallway, or contacts below

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Scan to connect on LinkedIn

Taboola